

If the PLC Cannot Read the Number Cleanly, the Flowmeter Project Is Not Finished

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Ruby L. | Velomac Flow Meter



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Some flowmeter projects look complete too early.

The meter is selected. The size is correct. The pressure and temperature are reviewed. The supplier sends the quotation.

Then startup comes closer and somebody asks a question that should have been settled at the beginning:

How will the control system actually receive and use this signal?

I think many avoidable flowmeter problems start there.

The instrument itself may be suitable. The line conditions may be acceptable. The technology choice may be reasonable.



Process Instrumentation

But if the PLC, DCS, or monitoring system cannot accept the signal path clearly, the project is still incomplete.

At that point, the site does not really have measurement yet.

It has hardware waiting for an argument.

That is why I prefer to ask integration questions much earlier in the process:

- Does the plant need 4–20 mA ?
- Is pulse output required?
- Is the system expecting RS485 , HART , Or Modbus ?
- Does the team need local display, remote display, or both?
- Who will actually use the number after startup?

Those are not paperwork questions.



They shape whether the meter becomes useful in daily operation.

Across different Velomac product families, the brochure references output and communication options such as 4–20 mA , pulse, RS485 , HART , and Modbus depending on configuration. The exact output still has to be confirmed for the selected model, but that is exactly the point:

the output decision is part of the meter decision.

I do not separate them.

A conductive liquid application may narrow toward electromagnetic measurement. A steam, gas, or liquid application with robust industrial conditions may keep vortex in the conversation. Direct gas mass flow may point toward thermal mass. A difficult retrofit may reopen the discussion around ultrasonic, V-Cone, or balanced differential pressure depending on installation reality.



Process Instrumentation

But even after the measurement principle is narrowed, one question remains:

Can the plant actually use the number without confusion?

If the answer is unclear, I would slow down before the order is approved.

Because startup problems are expensive in a different way. They do not always look dramatic on paper, but they waste engineering time, delay trust, and make the meter look guilty before the system around it is ready.

I also think calibration should stay in the same conversation. If the site wants the number for utility review, internal balancing, process control, or operational reporting, the team needs confidence in more than the hardware.

Velomac has 100% in-house calibration together with traceable gas and liquid calibration capability because believable measurement does not start at the terminal block. It starts with how the device was selected, verified, and integrated.

When a plant later says, “The flowmeter is not working,” sometimes what it really means is:

the signal expectation was never defined carefully enough.

That is why I do not treat communication details as an accessory.

They are part of the job.

If the PLC cannot read the number cleanly, the flowmeter project is not finished.

Flowmeter Signal +

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Overseas executive at Velomac Flow Meter. Practical notes on flow measurement, selection, steam, gas, liquids and energy loss blind spots.

